

Profile your Linux system – the baseline

Session 2 – Linux & cloud infrastructure foundations

Exercise 2 · Session 2 lab · Estimated time: 60–90 min

Write your answers in the answer document `SKY2100_Exercise2_Answers.docx`. You cannot detect something *abnormal* until you know what *normal* looks like. Today you measure your VM across the four pillars (CPU, memory, disk, network) plus its logs, and record a **baseline** you will reuse all semester. These are the same instruments that later feed your IDS (Session 9) and your incident timeline (Session 10).

1 Measure the four pillars

▷ Install the monitors:

```
sudo apt update && sudo apt install -y htop btop
```

Note: If `btop` is unavailable on your release, use `htop`; if neither installs, `top` is always present. Record which one you used in the answer document.

▷ Run each command, read the output, and record the reading in the baseline table of the answer document (same five rows):

- CPU model & cores — `lscpu`
- Total / available RAM — `free -h`
- Disk total / free (/) — `df -h`
- Listening ports (count + which) — `ss -tulpn`
- Current load average — `top` (or `htop` / `btop`)

▷ Read the recent log as well:

`journalctl -xe | tail -n 30` — nothing to record, but you will need this command again in Sessions 9 and 10.

2 Cloud transfer: Microsoft Learn

▷ Complete “Describe the core architectural components of Azure”

(Part 2 of the series).

learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure

▷ Map the concepts.

In the answer document, write one line each connecting the training to the lecture: Region, Availability Zone, Resource group, Subscription, Azure Resource Manager (ARM).

3 Azure reflection

▷ Your VM has a fixed CPU/RAM/disk. How does an Azure *region* and *availability zone* change the availability picture compared with your single local machine?

▷ `ss -tulpn` shows the ports your VM exposes. In Azure, which construct would limit who can reach those ports, and at which level (host vs. network)?

Key idea: The numbers you just wrote down are not interesting on their own. They matter because they are what “normal” looks like on your machine, and you can only recognise trouble by comparing against them. When Session 9 has your intrusion detection flag a process, or Session 10 asks you to explain a spike in a log, the first question is always the same: is this different from the baseline? An unknown port in **ss**, a load average far above today’s, or free memory near zero all mean something only because you measured the quiet state first. Keep this baseline; you will compare against it for the rest of the course.

4 Knowledge check

1. Which command lists the ports your machine is *listening* on?

☐ **free -h** ☐ **ss -tulpn** ☐ **df -h** ☐ **lscpu**

Hint: The socket-statistics command you ran for the baseline today.

2. State the difference between a **Type 1** and a **Type 2** hypervisor, with one example of each.

Hint: One runs directly on the hardware, one runs as an app on a host OS.

3. In a data centre, **east-west** traffic means...

☐ client ↔ data centre ☐ server ↔ server inside the data centre ☐ data centre ↔ internet

Hint: Most data-centre traffic stays server-to-server, inside the building.

4. Comer notes that running several VMs on one server adds overhead. Give *two* sources of that overhead.

Hint: Each VM carries its own OS — its own scheduler, background processes, memory.

5. An **Availability Zone** is best described as...

☐ a whole region ☐ a physically separate data centre within a region ☐ a resource group
☐ a subscription

Hint: A physically separate data centre inside one region.

6. Put the Azure hierarchy in order from *broadest* to *narrowest*: resource, management group, subscription, resource group.

Hint: From the top down: management group, then narrower, to the single resource.

7. Why is a virtual machine described as “a digital object”, and what is *one* security risk that follows from it?

Hint: Its entire state is data — which means it can be saved to a file, and copied.

8. In **free -h**, which value best indicates memory *pressure*?

☐ total ☐ available / swap in use ☐ buffers ☐ shared

Hint: Not the ‘free’ column — the one that counts cache as usable.

9. Explain **hot-aisle** / **cold-aisle** containment in one sentence, and say which CIA property it ultimately protects.

Hint: It is about airflow: keep cold intake and hot exhaust apart — which property does overheating threaten?

10. What is a **hypervisor**, and which CPU feature lets it stay more privileged than the guest operating systems?

Hint: It creates and mediates the VMs; CPU rings let it outrank the guests.

11. Name the three persistent-storage abstractions used in cloud data centres (Comer Ch. 8).

Hint: Block, file, and the key-value one.

12. (Reflection) Which single base-metric command will you rely on most when hunting an incident, and why?

Progress check: Discuss your answers in the group, then talk the teacher through them verbally at the next session

only to track progress; nothing is handed in, signed or graded.

Solutions

Work through the lab first, then check here. Model answers are indicative — equivalent reasoning is fine.

Note: Model answers are indicative – accept equivalent reasoning. Multiple-choice answers are in **bold**.

Knowledge check

1. **ss -tulpn**.
2. **Type 1** (bare-metal) runs directly on hardware – ESXi, Xen, Hyper-V; **Type 2** (hosted) runs as an app on a host OS – VMware Workstation, VirtualBox.
3. **Server ↔ server inside the data centre**.
4. Each VM runs its *own* OS, so each adds a *scheduler* and a set of *background processes*; plus per-VM memory/CPU and the time to boot an OS. (Comer Ch. 6, p. 71–72, verified.)
5. **A physically separate data centre within a region**.
6. **Management group → subscription → resource group → resource**.
7. A VM’s entire state (memory, disk, devices) is data, so it can be saved as a file/snapshot; risk: it can be *copied or stolen wholesale*, and a snapshot may capture secrets in memory. (Comer Ch. 5, p. 63, verified.)
8. **Available memory / swap in use**.
9. Racks are arranged so *cold intake* and *hot exhaust* air never mix, improving cooling efficiency; it ultimately protects **Availability** (keeps hardware running). (Comer p. 40, verified.)
10. A **hypervisor** creates and mediates VMs; CPU **privilege levels (rings)** let it run more privileged than the demoted guest OSes. (Comer p. 59–60, verified.)
11. **Block, file, and object (key-value) storage**. (Comer Ch. 8, verified.)
12. Reflection – commonly `journalctl` (the log source for the timeline) or **ss** (attack surface).

References

Comer Ch. 4 (hot/cold aisles **p. 40**, east-west traffic **p. 44**, leaf-spine **p. 47**) & Ch. 5 (approaches **p. 55**, privilege levels **p. 59**, extending privilege **p. 60**, levels of trust **p. 60**, VM as digital object **p. 63**, VM migration **p. 64**, live migration **p. 65**); Ch. 6 VM overhead **p. 71**; Ch. 8 storage abstractions – all verified. Jøsang Ch. 3 (syllabus, page numbers not verified). MS Learn: *Describe the core architectural components of Azure* (verified, June 2026).